LAB 9: Wireless Network Setup and Packet Analysis using Cisco Packet Tracer

1. Objectives

- Configure a wireless router and establish a fully functional wireless LAN using Cisco Packet Tracer.
- Connect wireless client devices and assign IP addresses automatically using DHCP.
- Verify secure wireless communication using WPA2-Personal security with AES encryption.
- Analyze network packets to understand DHCP, ARP, and data flow in a wireless environment.

2. Theory

2.1 Wireless Networks (WLAN)

A Wireless Local Area Network (WLAN) allows devices to communicate without physical cables using radio frequency signals, typically in the 2.4 GHz or 5 GHz bands. WLANs follow the IEEE 802.11 family of standards widely known as Wi-Fi. In infrastructure mode used in this lab all wireless clients communicate through a central wireless router rather than directly with one another. The router acts as the central access point, DHCP server, and gateway for the connected laptops.

2.2 SSID (Service Set Identifier)

The SSID is a name string of up to 32 characters that identifies a specific wireless network. Access points broadcast the SSID periodically in beacon frames which allow wireless devices to discover and list available networks. When a client selects a network and connects it uses the SSID to identify the correct access point. If two access points share the same SSID clients can roam between them seamlessly which is the basis of enterprise wireless deployments.

2.3 WPA2 Security and AES Encryption

WPA2 (Wi-Fi Protected Access 2) is defined by the IEEE 802.11i standard and is the industry standard for securing wireless networks. It replaced the older WEP and WPA protocols which had well-documented vulnerabilities. WPA2-Personal also known as WPA2-PSK uses a Pre-Shared Key which is a passphrase configured on both the router and the client to authenticate the device and generate session encryption keys.

WPA2 uses AES (Advanced Encryption Standard) combined with CCMP (Counter Mode CBC-MAC Protocol) which provides both data confidentiality and message integrity. AES is a symmetric block cipher operating on 128-bit data blocks making WPA2 highly resistant to eavesdropping and unauthorized access.

2.4 DHCP in Wireless Networks

Wireless routers include a built-in DHCP server. When a laptop joins the wireless network it uses the DORA process (Discover, Offer, Request, Acknowledge) to receive an IP address, subnet mask, default gateway, and DNS server address from the router. This is especially important in wireless environments where many devices frequently connect and disconnect making manual IP configuration impractical.

2.5 Packet Analysis in Wireless Networks

Packet analysis is the process of capturing and examining network packets to understand communication between devices. In Cisco Packet Tracer the Simulation mode allows students to step through each packet and inspect its contents. Key packet types in a wireless setup include DHCP packets during address assignment, ARP packets when resolving IP to MAC addresses, and ICMP packets during ping tests. This helps verify correct operation and assists in diagnosing configuration problems.

3. Network Topology

The topology for this lab consists of one Linksys WRT300N Wireless Router with two laptops connected wirelessly: Laptop0 and Laptop1. Both laptops have a Wireless-N notebook adapter (WPC300N) installed. No wired client connections are used in this lab.

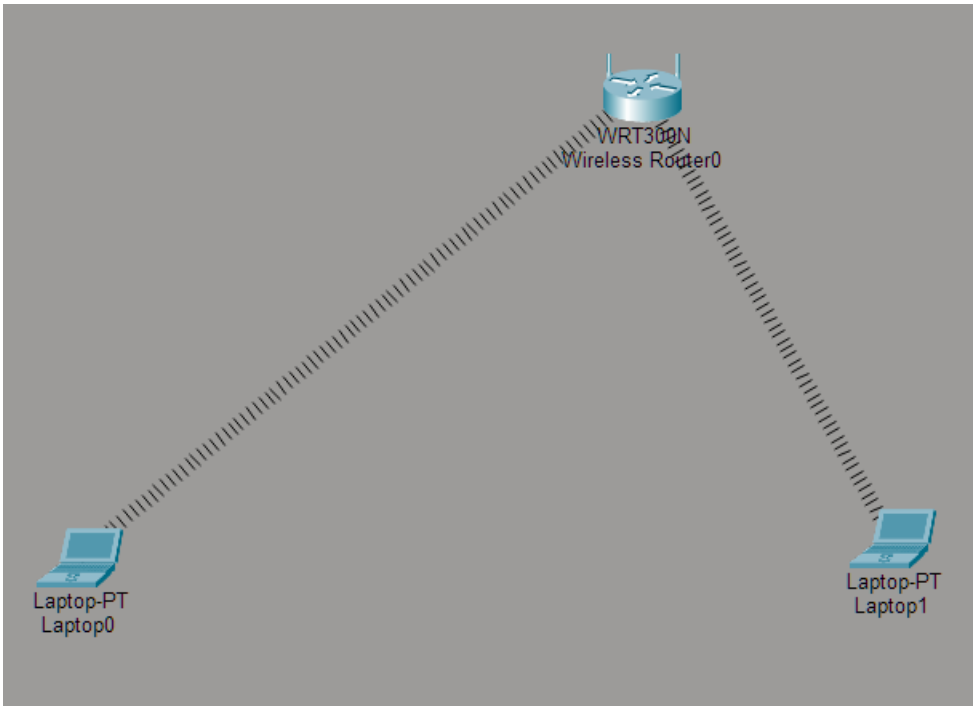

Figure 1: Network Topology showing WRT300N Wireless Router connected wirelessly to Laptop0 and Laptop1

4. Configuration Summary

Parameter	Setting	Parameter	Setting
Router IP	192.168.0.1	SSID	HCOE
Subnet Mask	255.255.255.0	Security Mode	WPA2-Personal
DHCP Server	Enabled	Encryption	AES
Network Mode	Mixed (B/G/N)	Passphrase	hcoe@123

5. Configuration Procedure

5.1 Step 1: Basic Router Setup

1. Click on the Wireless Router and open the GUI tab then navigate to Setup.
2. Set the Router IP Address to 192.168.0.1 and Subnet Mask to 255.255.255.0.
3. Enable the DHCP Server so it assigns IP addresses to connecting clients automatically.
4. Save settings.

5.2 Step 2: Wireless Network Configuration

5. Navigate to Wireless then Basic Wireless Settings.
6. Set Network Mode to Mixed, SSID to HCOE, and Channel to Auto.
7. Click Save Settings.

5.3 Step 3: Wireless Security

8. Navigate to Wireless then Wireless Security.
9. Set Security Mode to WPA2-Personal, Encryption to AES, and Passphrase to hcoe@123.
10. Save the configuration.

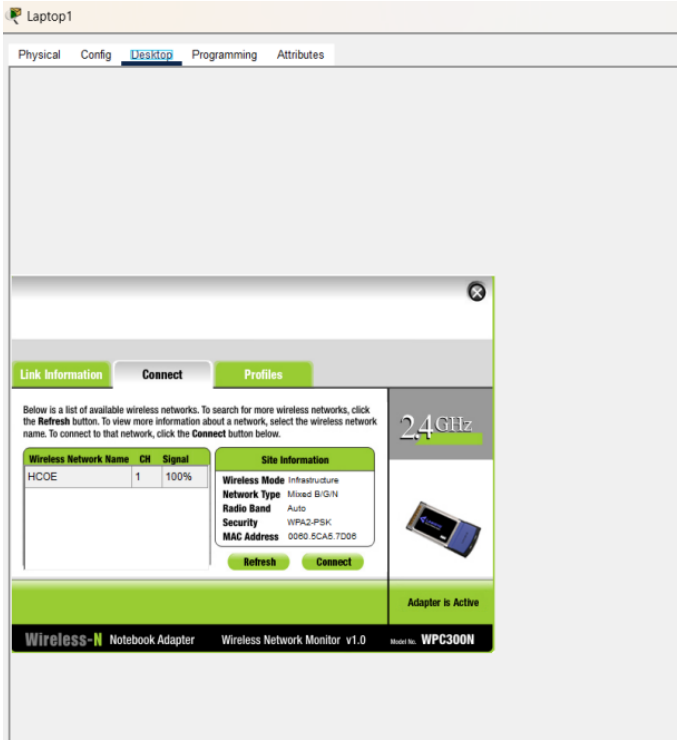

Figure 2: Laptop1 wireless adapter detecting the HCOE network with WPA2-PSK security at 100% signal strength

5.4 Step 4: Connecting Laptops to the Network

11. On each laptop verify the Wireless-N adapter (WPC300N) is installed via the Physical tab.
12. Open Desktop then PC Wireless.
13. Click the Connect tab, select the HCOE network from the list, and click Connect.
14. Enter the WPA2 passphrase hcoe@123 when prompted.
15. The laptop completes authentication and receives an IP address from the DHCP server automatically.

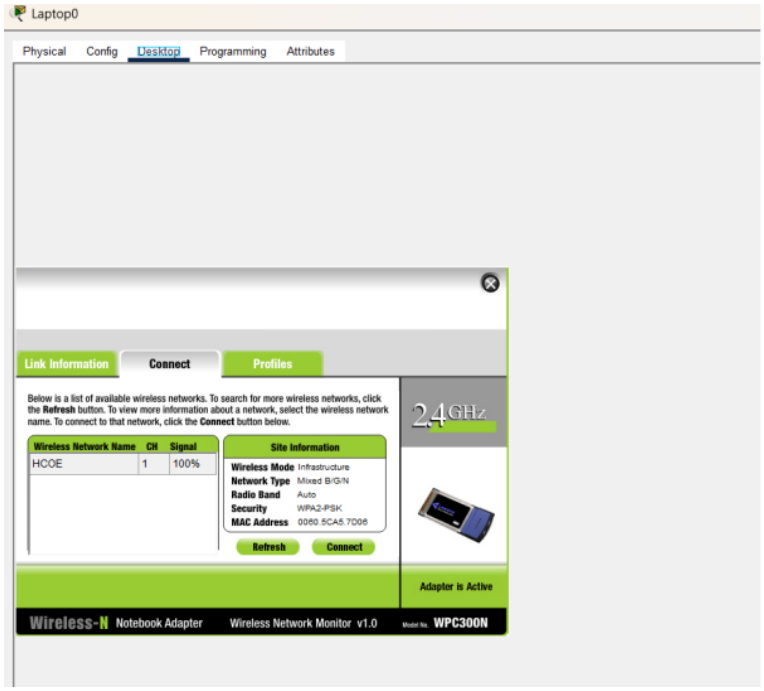

Figure 3: Laptop0 detecting the HCOE network at 100% signal strength and ready to connect

6. Result

The wireless network was established successfully. Both Laptop0 and Laptop1 detected the HCOE network on Channel 1 at 100% signal strength. After entering the WPA2-PSK passphrase both devices were authenticated and received valid IP addresses from the router DHCP server. Connectivity was confirmed and data packets were exchanged successfully between both laptops and the router.

7. Discussion and Conclusion

This lab demonstrated the complete process of setting up a secure wireless network. Configuring the SSID made the network discoverable, WPA2-Personal with AES encryption restricted access to authorized devices only, and the DHCP server handled automatic IP configuration for both laptops without any manual intervention.

From a packet analysis perspective the lab confirmed that the same core protocols used in wired networks also operate in wireless environments. ARP resolves IP addresses to MAC addresses, DHCP handles automatic configuration, and ICMP verifies connectivity. The additional steps specific to wireless are WPA2 authentication and the four-way handshake that takes place before normal IP traffic is allowed to flow.

The lab was completed successfully. A functional and secure wireless LAN was established, both client laptops connected and received IP configuration automatically, and wireless communication through the router was fully verified.